

System Design

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LLD (Low-Level Design)

* What is LLD?

- While DSA focus on solving isolated problems (like searching or sorting). LLD is the process of building a complete, real-world applications around those solutions.
- It involves defining the objects, entities, and relationships between them.
- LLD is essential for creating scalable, maintainable and robust software systems.

* LLD Vs. DSA :-

For a Ride-Booking App

- The DSA approach → Focuses solely on finding algorithmic solutions.
 - Graphs for finding shortest paths.
 - Priority Queue for mapping closest drivers to users.
 - Limitations: Fails to account for object interactions, data security, notification systems, and scaling.
- The LLD approach → Focuses on the system's architecture first.
 - Identify key objects: Users, Drivers, Locations, Notifications, Payments.

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- Establishes relationships and interactions between these objects.
- Addresses ~~and~~ non-functional requirements like data security and scalability.

* Pillars of LLD:-

- Scalability → How the application handles a surge in traffic (millions of users) without breaking.
- Maintainability → Writing code that is easy to debug and allows for new features to be added without breaking existing ones.
- Reusability → Using a "plug-and-play". Components like payment gateways or notification systems should be decoupled from the core application logic so they can be reused across different projects.

* What LLD is Not (HLD):-

While LLD focuses on code structure and object interactions, HLD concerns itself with the system architecture. It addresses:

- Tech Stack → Choosing the language and framework (e.g. Java, Spring Boot).

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Sagar OKA

- Database Selection → Deciding between SQL, NoSQL, or hybrid models.
- Server Scaling → Planning how the system scale servers to handle bulk traffic.
- Cost Optimization → Managing infrastructure expenses to ensure efficiency.

* Relationship between DSA, LLD & HLD :-

These three components form the complete foundation for building a robust application:

- DSA (The Brain) → Provides the fundamental algorithms and data structures to solve specific computational problems.
- LLD (The Skeleton) → Defines the internal code structure, class hierarchies, and object interactions.
- HLD (System Architecture) → Determines the overall system blueprint, including servers, databases & high-level tech choices.

If DSA is the brain, LLD is the skeleton of your application